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Research article

Anthropogenic forcing exacerbating the urban heat islands in India

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ABSTRACT

Urban heat island (UHI) phenomena is among the major consequences of the alteration of earth's surface due to human activities. The relatively warmer temperatures in urban areas compared to suburban areas (i.e. UHI) has potential health hazards, such as mortality due to high temperatures and heat waves. In addition, UHI situation demands more energy (e.g. fans and air-conditioners) that would trigger greenhouse gas emissions. Studies on UHI intensity help to assess its impact on urban population, city planning, and urban health planning. This is particularly important for a country like India, where 32% people (~7% of total world population), live in urban areas. We conducted a detailed study on surface UHI intensity (SUHII), which is the difference between urban and surrounding rural land surface temperatures, across all seasons in 44 major cities of India, which shows that mean daytime SUHII is positive (up to 2 °C) for most cities, as analysed from satellite temperature measurements for the period 2000–2017, in contrast to previous studies. However, although statistically insignificant, most cities show a positive trend in SUHII for monsoon and post-monsoon periods, but negative for winter and summer seasons. The increasing night-time SUHII in all seasons for most cities suggest increasing trend in temperature in cities due to the impact of the rapid urbanisation, and thus, suggesting the influence of anthropogenic forcing on SUHII. This is also supported by the analysis of aerosols, night lights, precipitation and vegetation in the study regions. Therefore, this study shall aid planning and management of urban areas by giving insights about the effects of nature and intensity of development, land cover and land use mix and the structure of cities on SUHII.

1. Introduction

More than half of the population in the world lives in urban areas. Due to the rapid urbanisation and growth of the population, about 2.5 billion people will be added to the urban areas by 2050, with India, China and Nigeria together accounting for about 35% of the urban population between 2018 and 2050 (UN, 2018). The process of rapid urban development leads to elevated temperatures in urban regions compared to their neighboring suburban or rural areas. This phenomenon is called the urban heat island (UHI) (Oke, 1973). Developments in the field of thermal infrared (TIR) remote sensing have significantly improved the study of surface UHI based on the land surface temperature (LST), since these datasets provide better spatial coverage, covering the entire urban area at the same time, compared to in-situ data which are point measurements (Cai et al., 2011; Rigo et al., 2006; Stewart and Oke, 2012; Voogt and Oke, 2003). Surface urban heat island intensity (SUHII) is defined as the difference between the LST of urban and its surrounding non-urban area.

Several studies have been performed to examine SUHII of different cities around the world (Clinton and Gong, 2013; Imhoff et al., 2010; Jin, 2012; Peng et al., 2012; Zhang et al., 2010). For instance, an analysis for more than 400 big cities in the world reported an annual daytime SUHII of 1.5 °C and night-time SUHII of 1.1 °C (Peng et al., 2012). A positive correlation between precipitation and population with annual mean day and night UHI, respectively, was found in a study based on 65 cities in the United States of America and Canada (Zhao et al., 2014). The UHI analysis for the 5000 largest cities in Europe showed that SUHII increases with size of the cities and their fractal dimension, but diminishes with the logarithm of the anisometry (Zhou et al., 2017). The seasonal and diurnal variation of SUHII of Asian megacities showed significant positive SUHII, in both magnitude and extent to which the surface UHI was present in 2001 (Tran et al., 2006). Also, a study of 32 cities in China showed an annual mean SUHII up to 1.85–1.95 °C (Zhou et al., 2014). A similar study of various cities across the world has showed the fact that the urban areas are warmer compared to their neighboring rural areas and the intensity of SUHII varies with their

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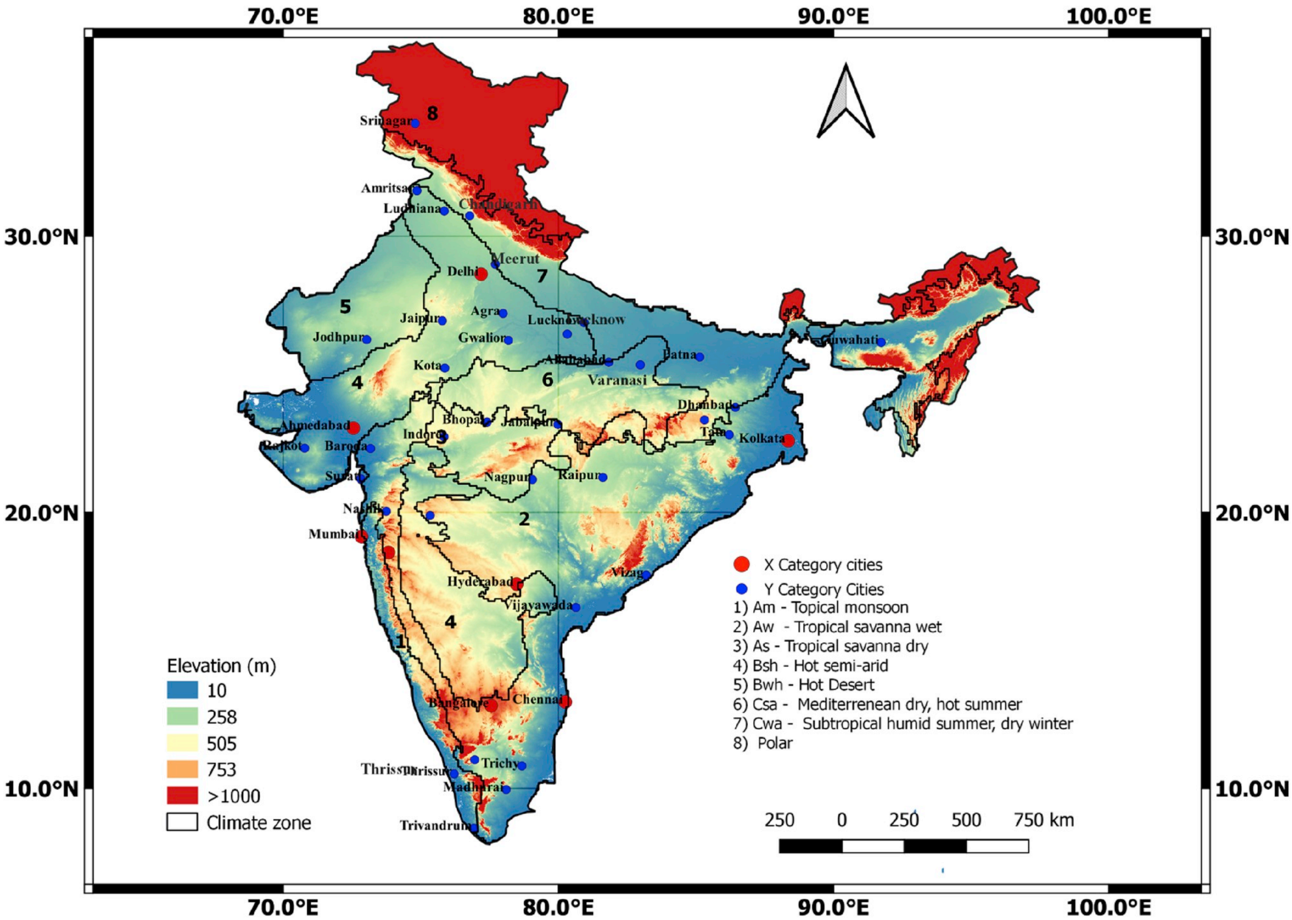

Fig. 1. Major cities of India. Geographical location of the cities selected for this study and digital elevation map of India. The red dots represent the cities classified as X category, which has a population greater than 5 million and blue dots represent cities classified as Y category that has population less than 5 million. The black lines represent the Koppen climate regions and the numbers signify the corresponding Koppen climate class. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

geographical location, size, population, and vegetation cover (Clinton and Gong, 2013). Doick et al. (2014) and dos Santos et al. (2017) showed that green cover in urban areas plays a significant role in UHI mitigation. A study of European cities showed that the cities in colder climatic region and cities with higher green cover are more prone to heat waves (Ward et al., 2016). A recent study also investigated the relationship between surface UHI and land cover types, and concluded that the SUHII was significantly modified by the built-up and vegetation land cover classes (Yang et al., 2017). The intensity and spatial extent of surface UHI in China is increasing over the years and this trend will continue if mitigation strategies are not put in place(Peng et al., 2018; Yang et al., 2019; Yao et al., 2019, 2018, 2017). A study of South American cities concluded the existence of significant positive SUHII both during day and night and also the influence of land cover on SUHII (Wu et al., 2019).

India has witnessed an increasing trend in urbanisation since its independence in 1947, and the economic liberalization after the 1990s has further amplified this trend. According to the 2011 census, 31.8% of the Indian population lives in urban areas, a 4% increase since the year 2001. Furthermore, India is expected to add another 416 million inhabitants to its cities by 2050 and leads the list ahead of China and Nigeria (UN, 2018). Subsequent infrastructure development is necessary to meet the needs of growing population in urban areas, which usually comes at the expense of conversion of vegetated areas and water bodies in the peri-urban areas to urbanised regions having impervious surfaces.

A recent study on 84 cities in India found negative SUHII during summer days and was attributed to the low vegetation cover in the adjacent rural region for the period 2003–2013 (Kumar et al., 2017; Shastri et al., 2017, 2015). There are also studies of UHI of individual cities. For example, a study of Delhi for years 2000 and 2010 reported that the increase in urban infrastructure has direct effect on LST and heat fluxes including anthropogenic heat flux (Chakraborty et al., 2015). Average annual SUHII of Chandigarh was shown to be about 4.98–5.43 °C and overall average SUHII has been observed to be 5.2 °C (Mathew et al., 2016). The UHI analyses conducted for Jaipur during the period 2003–2015 showed that significant surface UHI exists there with an average intensity of 7.86 °C (Mathew et al., 2017). A negative daytime SUHII was observed in Jaipur and a very weak day time SUHII was observed in Ahmedabad, while strong night-time SUHII was observed over both cities during the period 2003–2015 (Mathew et al., 2018).

Although there are UHI analyses for single (Chakraborty et al., 2015; Mathew et al., 2018, 2017, 2016) or multiple cities of India together (Kumar et al., 2017; Shastri et al., 2017, 2015), those discussions were mostly done for the connection between SUHII and vegetation or urban precipitation. However, many factors influence SUHII and they need to be analysed together for a better understanding of urban climate in the context of global warming induced by anthropogenic forcing (e.g. aerosols, intensity of anthropogenic activities). Therefore, a comprehensive analysis of 44 cities of India with a population more than 1 million for all seasons is performed. The spatial, diurnal as well as seasonal variations, trends in SUHII for the 2000–2017 period, and drivers of the spatial and temporal variability in SUHII are also discussed.

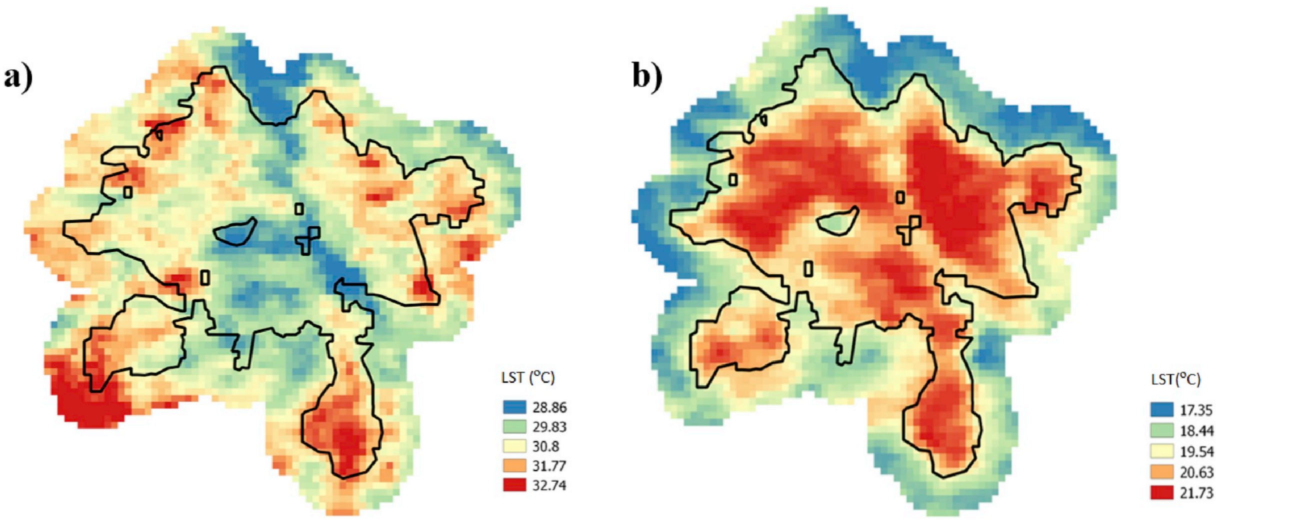

Fig. 2. Delineation of urban and rural boundaries for SUHII estimates. Annual mean land surface temperature of Delhi during daytime (a) and during night–time (b) in 2017. Also, the delineation of urban and rural areas is also depicted in the figure. Black line marks the urban boundary and the buffer zone around it considered as the rural region.

2. Data and methods

2.1. Datasets

Forty-four major cities across India are selected for this study. The cities are selected such that they fall into X and Y category cities as classified by the Government of India. X category cities have population greater than five million and Y category cities have population between five lakhs and five million. Furthermore, Y category cities which have population more than one million are only considered in this study. Small towns with population less than a million are excluded from this study, since the delineation of urban and rural areas is difficult in such urban areas, which would corrupt the SUHII analyses for big cities when averaged over the Indian or regional domain; leading to misinterpretation of the attribution of SUHII. Due to its vast size, the climate in India varies largely from region to region. India experience climate from four major Köppen climate groups and the cities belong to different Köppen climatic regions (Rubel and Kottek, 2010) (Fig. 1). LST was derived from the eight-day composite 1 km LST MODIS Terra (MOD11A2) V6 level-3 land 2017 (Wan et al., 2015). This dataset provides an average eight-day per-pixel LST. The MODIS bands 31 and 32 are used to derive LST data by using the generalized split-window algorithm, that corrects both atmospheric effects and surface emissivity. The MODIS V6 LST data are more accurate than previous versions and have errors within ± 1 °C (Wan, 2014). Each pixel value in the MOD11A2 is an average of all the corresponding daily MOD11A1 LST pixels collected within that eight-day period, which includes both day and night LST and the eight-day image composite removes the effects of cloud cover. The Terra satellite acquires data for India during day and night at around 11 a.m. (Indian Standard Time; IST) and 11 p.m. (IST), respectively. Only those data were selected in which at least 50% of the pixels were available without no data points. This will ensure the homogeneity of the analysis during cloud covered monsoon season. To remove the water pixels from the study area, the Terra MODIS Land/Water Mask (MOD44W) Version 6 data product, a global land/water mask derived from Terra MODIS is used. MODIS MCD12Q1 dataset that provides global land cover maps at annual time steps is used to delineate urban boundaries.

The dependence of SUHII with its probable drivers are also analysed using satellite derived measurements and following datasets that are collected for the period 2000–2017 are used for this purpose:

i) Enhanced vegetation index (EVI), which is a measure of vegetation, is derived from the MOD13A2 data product of the MODIS sensors aboard Terra satellite (K. Didan, 2015). The data are available every 16 days at 1 km spatial resolution as a gridded Level-3 product.
ii) The MODIS Terra and Aqua combined MCD19A2 Version 6 data product that provides land aerosol optical depth (AOD) gridded Level-2 product at 1 km pixel resolution is used to study the effect of aerosols on SUHII (Lyapustin and AuthorAnonymous, 2018).
iii) Night-time lights (NL) serves as a good proxy for measuring the anthropogenic activities in urban areas (Amaral et al., 2005; Elvidge et al., 2001). Satellite derived nightlights data are obtained from Defense Meteorological Satellite Program's Operational Linescan System (DMSP/OLS).
iv) The Tropical Rainfall Measuring Mission (TRMM) 3B43 monthly precipitation data at 0.25° resolution are used to characterise the effect of rainfall on SUHII (Precipitation Processing System (PPS) At NASA GSFC, 2018).

2.2. Analysis

SUHII is defined as the difference in mean LST between the urban areas and surrounding rural regions (Ward et al., 2016; Zhou et al., 2014). The urban areas are identified by applying the city clustering algorithm on MODIS MCD12Q1 annual international geosphere-biosphere programme (IGBP) classification scheme dataset that provides yearly global land cover types from 2001 to 2016 (Sulla-Menashe and Friedl, 2015). The rural areas are defined as a buffer area of all non-urban pixels around the city with an area approximately equal to the area of the city (Peng et al., 2012) (Fig. 2).

Trends are estimated using the annual aggregated time-series method and slope of the trend is estimated by linear least-squares regression of the annual means, and their significance at 0.05 level is estimated using the Mann-Kendall test (Forkel et al., 2013).

The LST derived from TIR measurements is controlled by radiative and thermodynamic properties of the surface. The exchange of energy between surface and atmosphere can be expressed as: $R_n = H + LE + \Delta S$, where R_n is the net radiation, H is the sensible heat flux, LE is the latent heat flux and ΔS is the surface heat flux (Arnfield, 2003; Voogt and Oke, 2003). With respect to surface UHI, the dominant fluxes in the energy balance equation during daytime are latent and sensible heat fluxes, while the storage surface heat fluxes become dominant during night.

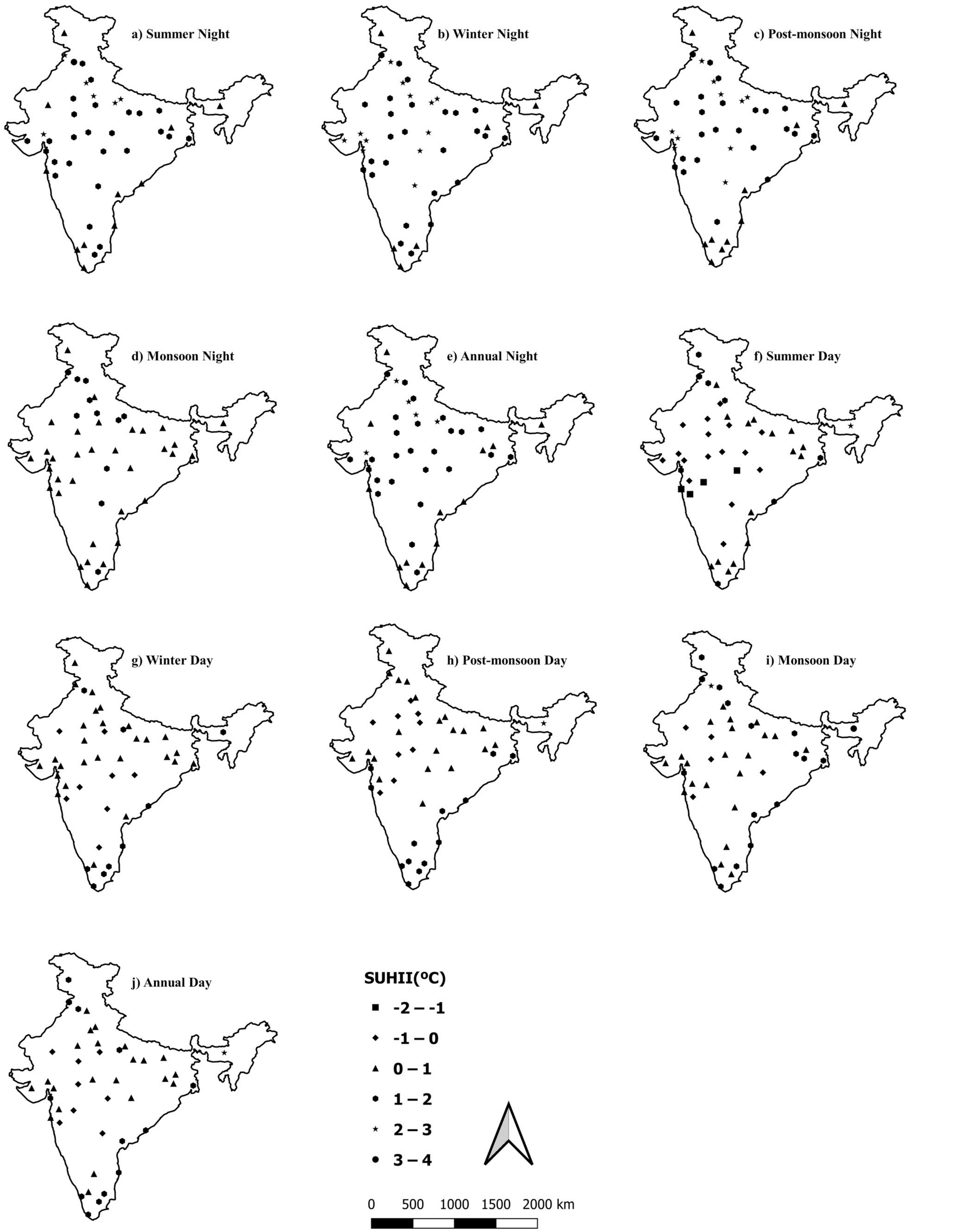

Fig. 3. The urban heat islands of India. The spatial distribution of surface urban heat island intensity (SUHII) for daytime and night-time across different seasons in 44 cities of India, including night-time SUHII of summer, winter, post-monsoon, monsoon and the annual average represented in **a, b, c, d,** and **e** respectively and the daytime counterparts in **f, g, h, i,** and **j** respectively.

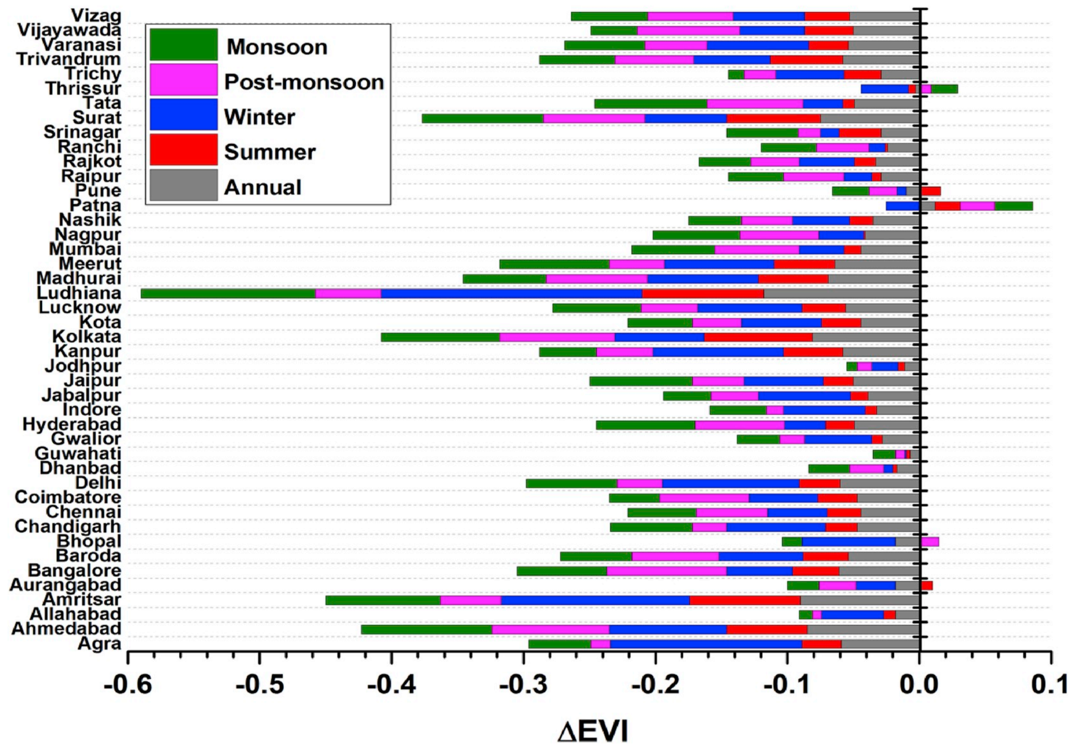

Fig. 4. The green cover in Indian cities relative to their rural proximity. The estimated difference in enhanced vegetation index (ΔEVI) of the 44 cities selected for the study. The bars are stacked and each colour bar represents the value of ΔEVI for the corresponding season. The ΔEVI values of summer, winter, post-monsoon and monsoon seasons are represented by red, blue, purple and green bars respectively. The grey bar represents the values of annual ΔEVI. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

These fluxes are also modified by surface types such as vegetation, impervious surfaces and water. The EVI, which is less influenced by atmospheric conditions, is used to measure the vegetated areas. The ΔEVI, which is the difference between EVI in the urban and suburban region is taken as a measure to quantify vegetation. NL data can capture the development activities in the urban areas and can be considered as a measure of anthropogenic activities in this city/urban area (Peng et al., 2012; Zhou et al., 2014). Therefore, ΔNL, which is the difference between NL data in the urban and suburban region is used as a proxy for analysing the effect of anthropogenic activities.

3. Results

3.1. The average SUHII in cities

The average SUHII of 44 Indian big cities are estimated and is depicted in Fig. 1. The cities are selected such that they represent at least 1 million people with significant built up area. In general, the daytime SUHII estimated show positive values in all seasons for most cities which implies that the urban areas are warmer than its surroundings. Out of 44 cities 20 cities show negative mean SUHII in summer and 9,8, and 5 cities in winter, monsoon and post-monsoon seasons, respectively. The mean SUHII varies between −0.9 °C and 1.5 °C in summer, −1.6 °C and 2.6 °C in winter, −0.8 °C and 2.0 °C in monsoon seasons and −0.6 °C and 1.7 °C in post monsoon season. However, six cities show negative SUHII in summer and winter, in which Jodhpur and Pune are common for all seasons. Aurangabad, Gwalior and Jodhpur are the other cities that show mean negative SUHII for all seasons, except in monsoon (Fig. 3). This is an expected result as the materials that constitute the urban built-up gets heated up more than surrounding rural areas. However, it is noteworthy that calculating the mean SUHII has its inherent limitations as it can be weighted by values of individual year or an anomalous SUHII occurrence in any year or season. Therefore, the SUHII averaged over the study period will not demonstrate the exact situation of the temporal evolution of SUHII and, hence the time series of SUHII in all seasons are also analysed (Fig. S1).

The monsoon and post-monsoon seasons show similar temporal evolution of mean SUHII as most cities exhibit positive SUHII from −0.75 °C to 3.7 °C (Fig. S1). Exceptions to these are the gradual increase of SUHII in Jodhpur from −0.62° to 2.12 °C, and the heavy rains and associated urban cool island in Varanasi in 2010 during monsoon (−2.51 °C). Agra, Delhi, Jaipur and, Jodhpur show negative SUHII for post-monsoon days. The night-time SUHII is positive for all seasons, from 0.33 °C to 2.35 °C, in all years, indicating that the urban surfaces emit more radiation than the surrounding rural regions during night. However, there are few exceptions such as 2016 for Srinagar in summer (−0.7 °C), 2001 for Ahmedabad in monsoon (−2.46 °C), and 2004 for Mumbai in post-monsoon seasons (−0.25 °C).

The daytime SUHII is highest for Guwahati, Chennai and Ludhiana (about 2 °C) and lowest for Pune (about 1 °C) for all seasons. This indicates that the SUHII is also related to the local weather/climate of the region, as the change in SUHII is analogous to the ambient temperature of the region (Zhao et al., 2014). However, the highest night-time SUHII (around 2.5 °C) is found for Ahmedabad and Ludhiana, and the lowest for Srinagar and Guwahati (0.3 °C), except during monsoon. The largest inter-annual variability is found in monsoon and the smallest in post-monsoon season. The large year-to-year SUHII change during monsoon leave no clear pattern for SUHII, which is directly connected to heavy or intermittent rainfall and evaporation during monsoon season. The results, therefore, show that about 85% of the cities have positive SUHII, with more or less similar values for each year, except for monsoon. This implies that the cities expand to their suburban areas and the temperature also increases there in tune with the changes in the adjacent cities. The rapid urbanisation of the type-Y (tier 2) cities in India is clearly identified in the SUHII analysis.

The night-time SUHII is positive for all seasons, indicating that the

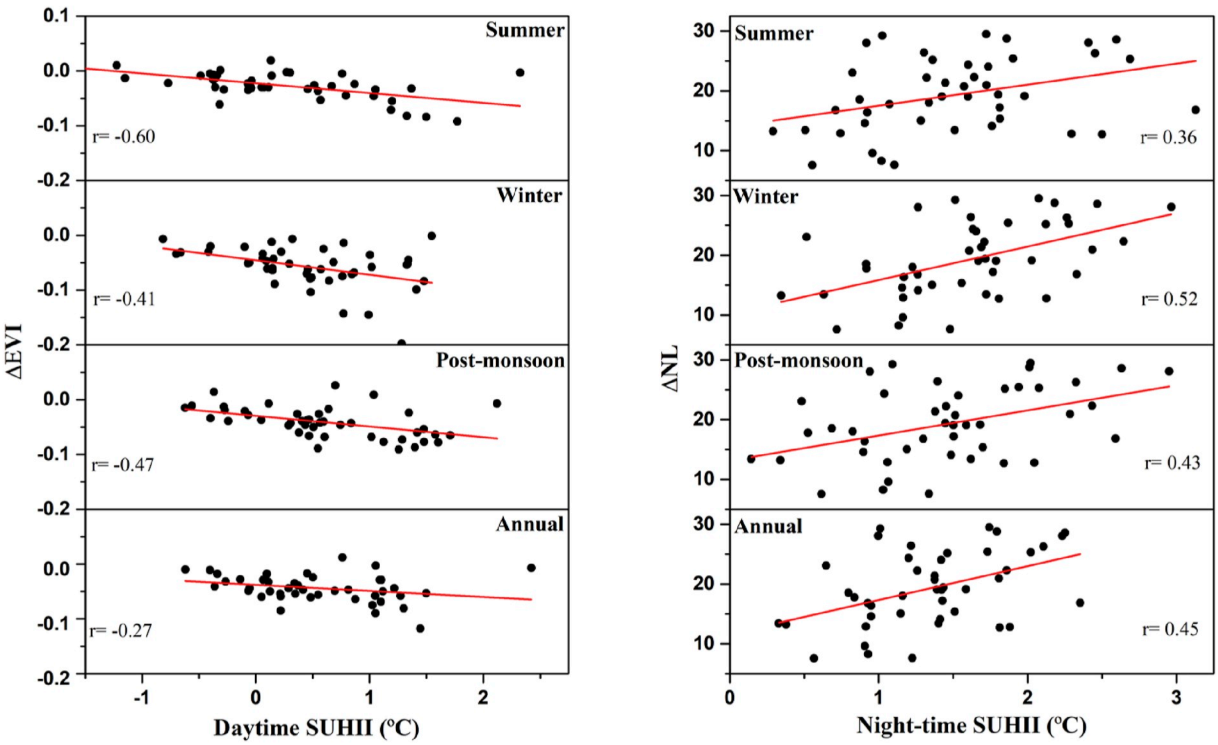

Fig. 5. Anthropogenic contributions to the heat islands. a) Dependence of daytime SUHII on ΔEVI and **b)** night-time SUHII on ΔNL. The left panel depicts the correlation of the daytime SUHII and ΔEVI and the right panel shows the correlation night-time SUHII and ΔNL, during different seasons. Corresponding r values are also specified in each panel.

night-time temperature is higher in all cities in all seasons and is in agreement with the previous studies (Kumar et al., 2017; Shastri et al., 2017). This consistent increase in night-time temperature suggests slow warming of urban and rural areas, and indicates the impact of global warming across the cities. This effect is also visible in daily temperature measurements, and is more apparent in big cities (Fig. S2, where built-up and intensity of development as well as the population density is significantly more).

Interestingly, our results are slightly different or even opposite to the conclusions drawn from previous studies (Kumar et al., 2017; Shastri et al., 2017), where they found negative SUHII in most Indian cities (mostly daytime SUHII) attributed to continuous decline in vegetation in rural areas of the corresponding. These results warranted a more detailed examination of SUHII in big Indian cities, and revealed that the SUHII average values were largely weighted by those of the small towns (42 cities in total), where city boundaries are hard to delineate and they house a population between 0.1 and 1 million. Nevertheless, the large cities with significant nightlights and distinct rural border still show the positive SUHII (e.g. Kolkata), consistent with the results of this study. It is further attested by the SUHII estimated for the cities in both studies with population of more than 5 million (e.g. Hyderabad and Coimbatore). Also, the cities that show contrasting results are also due to the difference in data analysis period (e.g. Lucknow) and in population (e.g. Dhanbad). Therefore, selection of cities and interpretation of SUHII must be done carefully for making conclusions on SUHII intensities, as addition of a year with more than average SUHII would decide the long-term average of SUHII in that city.

The results, nevertheless, for Jaipur and Ahmedabad are in agreement with the results reported in an earlier study conducted for the period 2000–2013 using MODIS data, although the values of SUHII are slightly different due to the difference in analysis period (Mathew et al., 2018). A study conducted for Chandigarh using the MODIS data for the 2000–2013 period show an average SUHII of 5°–6 °C for summer and winter seasons (Mathew et al., 2016) whereas results from this study show 0.74 °C/0.87 °C and 1.24 °C/1.76 °C for winter/summer day and night, respectively. This difference in SUHII estimates can be due to the difference in time and datasets used for the study.

4. Discussion

The SUHII exhibit different patterns for cities in the tropical, arid and temperate climate regions. The EVI, which is less influenced by atmospheric conditions, is used to measure the vegetated areas. The ΔEVI, which is the difference between EVI in the urban and suburban region is taken as a measure to quantify vegetation.

The large vegetated areas in suburban regions lead to higher evapotranspiration during day, which in turn leads to larger latent heat flux. This will cause an enhanced cooling effect on surface in suburban areas during day, as found for the higher SUHII in Kolkata during day, where a ΔEVI of −0.08 is observed (Fig. 4). In an arid area, the vegetation cover in both suburban and urban areas is smaller compared to a tropical city, and thus the effects of sensible heat flux dominate during day. Therefore, variability in skin temperatures of suburban and urban parts of these cities is smaller and that explains the reduced daytime SUHII. For example, Jaipur, located in a semi-arid area, has an annual average ΔEVI of −0.03 as the city is in the proximity of the Thar Desert. Conversely, the night-time temperature is controlled by surface heat fluxes. The urban built-up store more heat than the sparsely vegetated suburban areas of arid regions and thus shows higher SUHII at night. This is also reflected in the correlation between SUHII and EVI, because it shows statistically significant correlation for summer and winter SUHII during day ($r = -0.60, -0.41$, respectively Fig. 5).

A similar result was also found in a study conducted for 32 major cities of China (Zhou et al., 2014). Correlation between day and night-time SUHII is, however, very weak and statistically insignificant for all seasons. Since the interannual variation of UHI and EVI is very large during monsoon, there is no correlation between SUHII and EVI. This implies that EVI alone cannot explain the variability and spatial

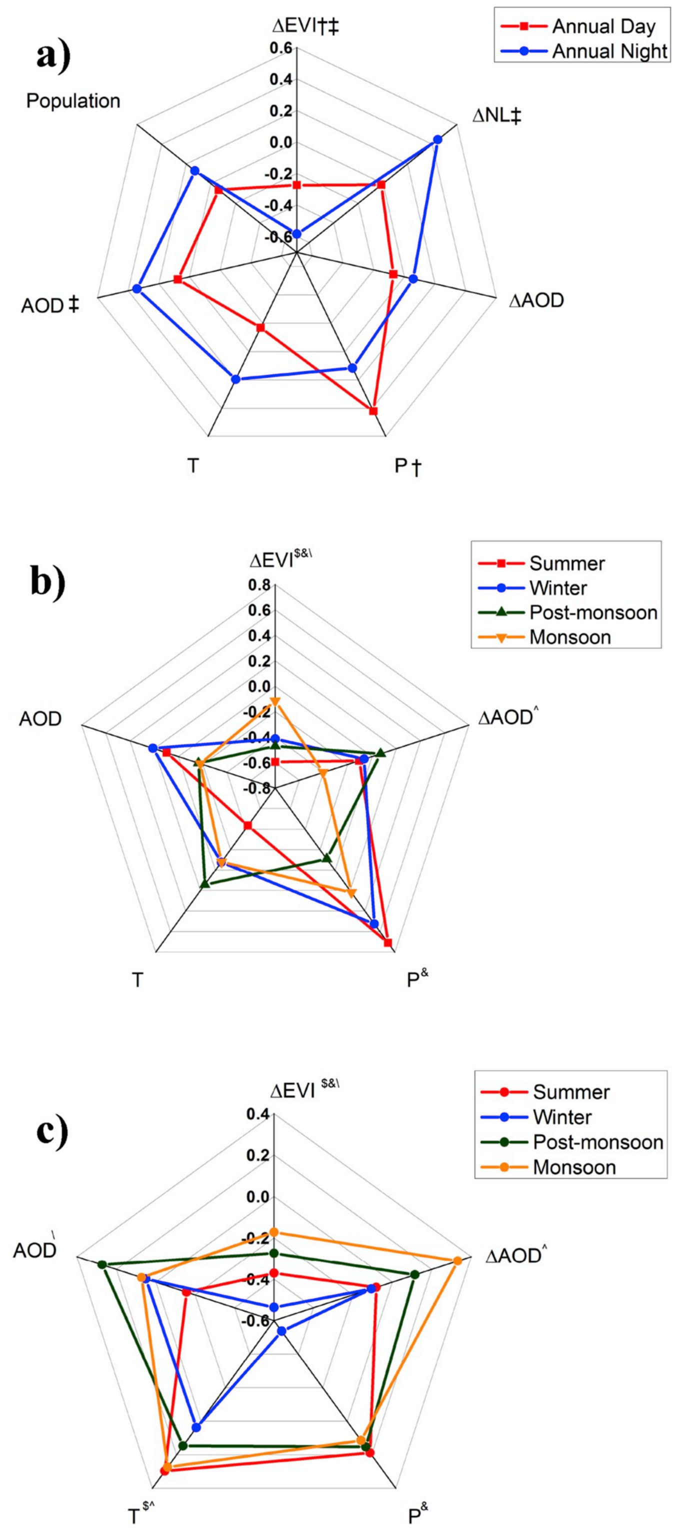

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Fig. 6. The nexus between urban heat islands and vegetation, atmospheric pollution, climate and socio-economic factors. Correlation of SUHII with ΔEVI, ΔNL, ΔAOD, mean precipitation (P), mean air temperature (T), mean aerosol optical depth (AOD) and population. **a)** Correlation of annual SUHII with ΔEVI, ΔNL, ΔAOD, mean precipitation (P), mean air temperature (T), mean aerosol optical depth (AOD) and population. Red line and blue line represent daytime and nigh-time SUHII, respectively. **b)** and **c)** represent correlation of daytime and night-time SUHII with ΔEVI, ΔNL, ΔAOD, mean precipitation (P), mean air temperature (T), mean aerosol optical depth (AOD). The summer, winter, post-monsoon and monsoon seasons are represented by red, blue, green and orange lines respectively. The symbols †, ‡, $, &, ¥, and ˆ represent statistically significant values during day, night, summer, winter, post-monsoon and winter seasons, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

inhomogeneity of SUHII. Therefore, the analysis of the night lights, aerosol loading, precipitation in the city and surrounding rural areas has also been carried out to further examine development of cities in terms of their built infrastructure and population.

The difference in nightlights between urban and suburban areas (ΔNL) is used to examine the effect of anthropogenic activities in nighttime SUHII (Amaral et al., 2005; Elvidge et al., 2001; Letu et al., 2010). Larger values of ΔNL indicate higher anthropogenic activities in urban areas, which in turn can lead to higher SUHII. Kolkata having a ΔNL of 27.7 has higher night-time SUHII during summer and winter (1.05 °C and 1.5 °C, respectively). However, Dhanbad, with a low ΔNL of 7.9, has a smaller night-time SUHII in both seasons (0.56 °C and 0.71 °C for summer and winter, respectively). The results show a statistically significant positive correlation between ΔNL and SUHII in the summer and winter seasons (Fig. 6b and **c).**

Another indicator of the anthropogenic forcing is the aerosol loading which is analysed using AOD data available from MODIS (Cao et al., 2016). The cities in the Indo-Gangetic plain exhibit the highest values of AOD (>0.45) across all seasons with larger AOD during the post-monsoon season (>0.5). The ΔAOD, which is the difference between the mean AOD's of the urban area and rural area is also analysed. The ΔAOD is negative for most cities (36 out of 44) during summer. In addition, 16 out of these 36 cities show negative daytime SUHII during this season and most exhibit smaller SUHII (<1 °C). However, Guwahati, located in the northeastern region of India, despite showing a high negative (−0.054) ΔAOD, experience a significantly higher SUHII (2.32 °C) during summer seasons. The ΔAOD was positive for most cities (32 out of 44) during the post-monsoon season, implying a higher aerosol loading over the urban areas during this season. The SUHII was also positive for most of the cities (35 out of 44) during the same season, although Guwahati exhibits higher SUHII (2.12 °C) albeit having ΔAOD of −0.014.

The precipitation data available for the study period was also analysed. The difference between urban and rural rainfall could not be estimated since the data available had a coarse resolution to carry out such an analysis, hence, the monthly rainfall averaged over the rural and urban areas were analysed. The mean precipitation showed a positive correlation with summer and winter ($r = 0.71$ and 0.52, respectively) SUHII during daytime and negative correlation (−0.52) with night-time winter SUHII (Fig. 6 b and c).

4.1. *Temporal evolution of SUHII*

The day and night SUHII trends for all seasons are estimated and the results are listed in Table 1. In general, the daytime summer and winter SUHII show negative trends for most cities; about 23 (summer) and 24 (winter) out of 44 cities, in which 10 and 13, respectively, are statistically significant. This implies that, although the mean SUHII is positive for most urban areas, the vegetation is decreasing and thereby temperature is increasing in city areas. A similar trend is also observed for

Table 1
The significance of urban heat islands. Trend (°C/decade) of daytime and night-time SUHII estimated for the period 2000–2017. Trends are shown with 95% confidence level. The statistically significant values are shown in bold.

No.	City	Summer		Winter		No.	City	Summer		Winter	
		Day	Night	Day	Night			Day	Night	Day	Night
1	Agra	**-0.278**	0.054	**-0.291**	-0.104	23	Kota	**-0.225**	-0.069	**-0.258**	**-0.209**
2	Ahmedabad	-0.124	**0.170**	**-0.093**	-0.096	24	Lucknow	0.007	-0.040	-0.065	**-0.264**
3	Allahabad	-0.114	**0.108**	-0.014	-0.130	25	Ludhiana	**-0.376**	-0.026	**-0.583**	0.019
4	Amritsar	**-0.368**	0.019	**-0.360**	0.042	26	Madhurai	-0.210	0.040	**-0.513**	0.058
5	Aurangabad	-0.176	**0.188**	0.022	**0.225**	27	Meerut	-0.022	0.059	-0.023	-0.045
6	Bangalore	-0.239	**0.117**	-0.130	0.127	28	Mumbai	-0.083	**0.093**	**0.124**	**0.177**
7	Baroda	**0.261**	**0.226**	**0.168**	0.024	29	Nagpur	-0.202	**0.109**	**-0.221**	**0.104**
8	Bhopal	-0.033	**0.196**	0.103	-0.010	30	Nashik	**-0.312**	**0.270**	**-0.107**	**0.307**
9.	Chandigarh	**-0.343**	0.014	**-0.241**	-0.007	31	Patna	**0.300**	**0.279**	0.044	0.099
10	Chennai	0.044	0.065	**0.310**	0.072	32	Pune	0.025	0.019	**0.180**	0.016
11	Coimbatore	0.124	**0.087**	-0.046	-0.013	33	Raipur	0.059	**0.116**	0.058	**0.132**
12	Delhi	**-0.313**	0.075	**-0.301**	-0.082	34	Rajkot	-0.214	**0.145**	0.227	0.088
13	Dhanbad	0.019	**0.047**	**0.062**	0.025	35	Ranchi	-0.097	**0.181**	-0.044	**0.146**
14	Guwahati	0.106	**0.245**	**-0.104**	**0.254**	36	Srinagar	**0.345**	**-0.158**	**0.458**	0.016
15	Gwalior	0.175	**0.181**	**0.146**	-0.044	37	Surat	0.008	0.082	**-0.173**	0.030
16	Hyderabad	-0.175	**0.238**	-0.115	**0.242**	38	Tata	0.156	**0.222**	**0.195**	**0.243**
17	Indore	0.058	**0.191**	**0.569**	0.062	39	Thrissur	**0.182**	**0.101**	**0.223**	0.001
18	Jabalpur	**0.372**	**0.328**	**0.078**	0.039	40	Trichy	**-0.351**	-0.025	**-0.311**	**0.104**
19	Jaipur	**-0.393**	0.015	**-0.355**	-0.37	41	Trivandrum	**0.427**	0.057	**0.434**	**0.184**
20	Jodhpur	0.093	**-0.182**	**0.126**	**-0.267**	42	Varanasi	0.010	**0.237**	0.028	0.030
21	Kanpur	-0.088	0.116	-0.077	**-0.192**	43	Vijayawada	0.174	**0.219**	-0.028	**0.109**
22	Kolkata	-0.030	**0.165**	-0.069	**0.270**	44	Vizag	**0.227**	0.053	**0.176**	**0.117**

The values in bold font represents statistical significance.

monsoon and post-monsoon seasons during daytime, as the trends of about 10 (monsoon) and 11 (post-monsoon) out of 44 cities show negative trends (−0.6° to −0.18 °C/decade), but only 1 city in monsoon and 2 cities in post-monsoon show (e.g. Amritsar and Ludhiana) statistically significant trends. These are also consistent with the temporal evolution of EVI, which is a measure of vegetation, during these seasons. The cities that show negative trend in SUHII has a negative trend in vegetation such as Chandigarh and Ahmedabad (−0.27° and −0.104 °C/decade, respectively). Similarly, a few cities show increasing trend in SUHII since the difference in vegetation in urban and suburban areas is increasing and Pune (0.18 °C/decade) is an example for such an urban agglomeration.

The trends in night SUHII show positive values for most cities (−0.26°–0.64 °C/decade), although more number of cities show negative SUHII during winter (15) and monsoon (18), and half of these for other seasons. As shown by the mean SUHII, the night-time SUHII shows positive trends (0.007°–0.64 °C/decade), indicating the continuous heating of urban surfaces in the night. The time for cooling the urban surfaces takes longer, indicating the slow and consistent increase of the background temperature of the city, which is a signature of anthropogenic warming or a component of the global warming. In addition, the night light differences between cities and their suburban areas are significantly reducing, which suggests the urban sprawl and increase of urban footprint. The conversion of suburban areas to urban agglomerations indicates development of the city too. In India, urbanisation is very rapid for many reasons such as migration of people for jobs, development of the cities to accommodate more built infrastructure for industry and other amenities. However, the difference in urban and rural nightlights is increasing in Nagpur and Mumbai, although there are also ambiguities in delineating urban and suburban areas in some complex megacities such as Mumbai.

Conservation and expansion of vegetated areas in and around cities could be an effective mitigation strategy to counter the effects of high SUHII. Evidence from this study suggest that more green spaces within the city's boundary could reduce the temperature in the city and surrounding areas. This comprehensive analysis of SUHII and its relationships between potential drivers across both space and time could aid in implementing accurate mitigation strategies of SUHII in urban areas in future.

5. Conclusion

This analysis shows that the SUHII was intense during day in cities in tropical climatic zones (Köppen class A) especially in cities of Chennai and Kolkata (1.21 °C and 1.33 °C) and during night in arid and temperate cities (Köppen class B and C) such as Delhi and Ahmedabad (2.25 °C and 2.23 °C). The SUHII is weak during night-time in the tropical cities such as Trivandrum and Visakhapatnam (0.8 °C and 0.91 °C), and during day in arid and temperate cities of Jaipur and Gwalior (−0.14 °C and 0.12 °C). The effect of vegetation on determining SUHII was evident as the lower vegetation cover in suburban areas of the arid and temperate climate regions was the major factor for a weaker daytime SUHII. There is an increasing trend in the night-time SUHII in all seasons for most cities (0.007°–0.64 °C per decade). This can be attributed to the effect of anthropogenic forcing, that play a major role in determining the night-time SUHII. The study therefore, demonstrates the effect of anthropogenic activities on the heat island of Indian cities, and the surface UHI effect will be enhanced if the city planning does not account for management of its green cover (e.g. EVI), pollution (e.g. AOD), local weather and climate (e.g. temperature and precipitation) and socio-economic factors (e.g. population). The findings of this study can be beneficial for implementing effective strategies to mitigate SUHII and its detrimental effects. Future research should be aimed at conducting in-depth analysis of the effects of SUHII and their drivers within individual cities.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2019.110006.

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